

Triadic Closure with Central Involution: A Minimal Generative Kernel for the Fano Plane and Octonionic Multiplication

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Abstract

We present a minimal structural kernel in which triality and a central involution are taken as primitive. A triangulation law is defined and shown to induce a closure operation whose minimal nontrivial finite realization is a seven-element incidence structure. The resulting closure system satisfies pairwise completion constraints that force the minimal size $n = 7$. The involution organizes this realization into a $3 + 1 + 3$ decomposition and conjugates orientation classes, yielding anti-phase symmetry. The resulting oriented closure system realizes octonionic multiplication. A scale-free recursive extension is stated.

1 Primitive Kernel

Definition 1 (Triality). *Let $\{X, Y, Z\}$ be three generators.*

Definition 2 (Central Involution). *Adjoin a distinguished element m and define*

$$\Phi(a) = -a, \quad \Phi(m) = m, \quad \Phi^2 = \text{id}.$$

2 Triangulation Law

Definition 3 (Triangulation Operator). *Let S be a set with distinguished center m . A triangulation operator is a map*

$$\mathcal{T} : S \times S \rightarrow S$$

such that for distinct $a, b \in S$, $\mathcal{T}(a, b)$ is the unique element $c \notin \{a, b\}$ for which the set $\{a, b, c\}$ is invariant under pairwise application of $\mathcal{T}$.

Definition 4 (Reciprocity). *The triangulation operator satisfies*

$$\mathcal{T}(a, b) = \Phi(\mathcal{T}(b, a)).$$

3 Closure Induced by Triangulation

Definition 5 (Closure). *Define the closure operation*

$$\mathcal{C}(a, b) := \mathcal{T}(a, b).$$

Proposition 1. *The closure operation satisfies*

$$\mathcal{C}(a, b) = \Phi(\mathcal{C}(b, a)).$$

Proof. By definition of $\mathcal{C}$ and reciprocity of $\mathcal{T}$. □

Remark 1. *Thus the triangulation operator induces a triadic closure system on S .*

4 Minimal Finite Closure

Definition 6 (Triadic Closure System). *A finite set S equipped with a closure operation $\mathcal{C}$ is said to form a triadic closure system if:*

1. *for all distinct $a, b \in S$, there exists a unique $c \in S$ with $c \notin \{a, b\}$ such that $\mathcal{C}(a, b) = c$;*
2. *$\mathcal{C}(a, b) = \Phi(\mathcal{C}(b, a))$;*
3. *each triple $\{a, b, c\}$ arising from $\mathcal{C}$ is closed under pairwise application of $\mathcal{C}$.*

Proposition 2 (Pair Counting Constraint). *Let $|S| = n$. Then*

$$\frac{n(n-1)}{2} \equiv 0 \pmod{3}.$$

Proof. Each unordered pair $\{a, b\}$ determines a unique third element $c = \mathcal{C}(a, b)$, forming a triad $\{a, b, c\}$. Each triad contains exactly three unordered pairs. Since every unordered pair appears in exactly one triad, the total number of unordered pairs must be divisible by 3. Hence

$$\frac{n(n-1)}{2} \equiv 0 \pmod{3}.$$

□

Proposition 3 (Minimal Nontrivial Size). *The smallest nontrivial solution of*

$$\frac{n(n-1)}{2} \equiv 0 \pmod{3}$$

is $n = 7$.

Proof. For $n = 3$, the system contains only a single triad and is therefore trivial. The values

$$n = 4, 5, 6$$

do not satisfy the divisibility condition. The next solution is $n = 7$, which is therefore the minimal nontrivial size. □

Proposition 4 (Structure of the Minimal System). *For $n = 7$, the triadic closure system consists of:*

- *7 elements,*
- *7 distinct triads,*
- *each unordered pair appearing in exactly one triad.*

Proof. The total number of unordered pairs is

$$\frac{7 \cdot 6}{2} = 21.$$

Since each triad accounts for 3 unordered pairs, the number of triads is

$$\frac{21}{3} = 7.$$

By the defining closure axiom, each unordered pair appears in exactly one triad. $\square$

Proposition 5 (Involution Decomposition). *The involution Φ partitions the minimal system into a $3 + 1 + 3$ decomposition.*

Proof. Since $\Phi^2 = \text{id}$ and $\Phi(m) = m$, all non-central elements occur in paired orbits

$$(a, \Phi(a)).$$

For $n = 7$, there are six non-central elements, hence three such pairs, together with the fixed central element m . $\square$

Theorem 1 (Uniqueness of the Minimal Closure Structure). *Any 7-element triadic closure system satisfying the axioms above is unique up to relabeling.*

Proof. By the preceding propositions, any such system consists of 7 elements arranged into 7 triads, with each unordered pair appearing in exactly one triad. This is precisely the defining incidence property of a Steiner triple system of order 7. It is a standard theorem that the Steiner triple system of order 7 is unique up to isomorphism. Hence any 7-element triadic closure system satisfying the stated axioms is unique up to relabeling. $\square$

Corollary 1. *The resulting incidence structure coincides with the Fano plane.*

5 Dyadic Realization

Define

$$D = \{1:7, 2:6, 3:5, 4:4, 5:3, 6:2, 7:1\}$$

with involution

$$\iota(a : b) = b : a.$$

6 Unified Mapping

$X \leftrightarrow 1:7 \leftrightarrow e_1$
$Y \leftrightarrow 2:6 \leftrightarrow e_2$
$Z \leftrightarrow 3:5 \leftrightarrow e_3$
$m \leftrightarrow 4:4 \leftrightarrow e_4$
$-Z \leftrightarrow 5:3 \leftrightarrow e_5$
$-Y \leftrightarrow 6:2 \leftrightarrow e_6$
$-X \leftrightarrow 7:1 \leftrightarrow e_7$

7 Orientation and Phase

Definition 7. For an oriented triple (a, b, c) , define

$$\Phi(a, b, c) := (\Phi(a), \Phi(b), \Phi(c)).$$

For an orientation class ω , define

$$\Phi(\omega) := \{\Phi(a, b, c) \mid (a, b, c) \in \omega\}.$$

Proposition 6.

$$\Phi(\omega) = -\omega.$$

Proof. Let $(a, b, c) \in \omega$. Then

$$\Phi(a, b, c) = (\Phi(a), \Phi(b), \Phi(c)) = (-a, -b, -c).$$

Since Φ acts by phase inversion on each generator while fixing the center, it sends each oriented triple to its conjugate orientation. Thus

$$(-a, -b, -c) \in -\omega,$$

so $\Phi(\omega) \subseteq -\omega$.

Applying Φ again and using $\Phi^2 = \text{id}$ yields

$$\Phi(-\omega) \subseteq \omega.$$

Applying Φ to the first inclusion gives

$$-\omega \subseteq \Phi(\omega).$$

Therefore

$$\Phi(\omega) = -\omega.$$

□

8 Octonionic Realization

Theorem 2. Octonionic multiplication is realized by the oriented closure:

$$e_a e_b = \pm e_c \quad \text{iff} \quad \mathcal{C}(a, b) = c.$$

9 Dyadic Multiplication Lines

$$\begin{aligned} (1:7, 2:6, 5:3), & \quad (2:6, 3:5, 7:1), & \quad (3:5, 1:7, 6:2), \\ (1:7, 7:1, 4:4), & \quad (2:6, 6:2, 4:4), & \quad (3:5, 5:3, 4:4), \\ & \quad (7:1, 6:2, 5:3). \end{aligned}$$

10 Recursion Principle

Definition 8. Define

$$\mathbb{T}_0 = \{X, Y, Z\}, \quad \mathbb{T}_1 \text{ as its closure.}$$

Statement 1. The system extends recursively under triangulation, preserving closure and involution. The explicit construction of the recursive lift is deferred.

11 Kernel Statement

Triangulation induces closure;
closure yields the unique minimal Fano structure;
oriented closure yields octonions.

12 One-Line Form

$$\mathbb{T}_0 \xrightarrow{\mathcal{T}} \mathcal{F}^\omega \xleftrightarrow{\Phi} \mathcal{F}^{-\omega}$$